SHODWE
Pizza Resto

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SQL PROJECT ON PIZZA SALES

WHERE EVERY SLICE TELLS A STORY

SQL PORTFOLIO

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PROJECT OVERVIEW

This project utilizes SQL by analyzing pizza sales data to gain insights into business performance. The analysis focuses on identifying total revenue, popular pizza categories, customer ordering patterns, and sales trends. By using SQL queries such as joins, aggregations, and filtering, the project converts raw sales data into meaningful insights that can help improve business decisions.

1. Retrieve the total number of orders placed.

```sql
SELECT
    COUNT(order_id) AS Total_Orders
FROM
    orders;
```

	Total_Orders
▶	21350

2.Calculate the total revenue generated from pizza sales.

```
SELECT
    ROUND(SUM(order_details.Quantity * pizzas.price),
            2) AS 'Total Revenue'
FROM
    order_details
        JOIN
    pizzas ON pizzas.pizza_id = order_details.pizza_id;
```

Total Revenue
817860.05

3. Identify the highest-priced pizzat

```
SELECT
    pizza_types.name, pizzas.price
FROM
    pizza_types
        JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
ORDER BY pizzas.price DESC
LIMIT 1;
```

name	price
The Greek Pizza	35.95

4. Identify the most common pizza size ordered.

```
SELECT
    pizzas.size,
    COUNT(order_details.order_details_id) AS Common_Pizza_Size_Order
FROM
    pizzas
        JOIN
    order_details ON pizzas.pizza_id = order_details.pizza_id
GROUP BY pizzas.size
ORDER BY Common_Pizza_Size_Order DESC;
```

size	Common_Pizza_Size_Order
L	18526
M	15385
S	14137
XL	544
XXL	28

14137

5. List the top 5 most ordered pizza types along with their quantities.

```
SELECT
    pizza_types.name,
    SUM(order_details.Quantity) AS Total_Ordered
FROM
    pizza_types
        JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
        JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY Total_Ordered DESC
LIMIT 5;
```

name	Total_Ordered
The Classic Deluxe Pizza	2453
The Barbecue Chicken Pizza	2432
The Hawaiian Pizza	2422
The Pepperoni Pizza	2418
The Thai Chicken Pizza	2371

6. Join the necessary tables to find the total quantity of each pizza category ordered.

About

```
SELECT
    pizza_types.category,
    SUM(order_details.Quantity) AS Total_Quantity
FROM
    pizza_types
        JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
        JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY Total_Quantity DESC;
```

category	Total_Quantity
Classic	14888
Supreme	11987
Veggie	11649
Chicken	11050

7. Determine the distribution of orders by hour of the day.

```
SELECT
    HOUR(order_time) AS Hours, COUNT(order_id) AS Total_Order
FROM
    orders
GROUP BY HOUR(order_time);
```

Hours	Total_Order
11	1231
12	2520
13	2455
14	1472
15	1468
16	1920
17	2336
18	2399
19	2009
20	1642
21	1198
22	663
23	28
10	8
9	1

8.Join relevant tables to find the category-wise distribution of pizzas.

```
SELECT
    category, COUNT(name) AS Total_items
FROM
    pizza_types
GROUP BY Category;
```

category	Total_items
Chicken	6
Classic	8
Supreme	9
Veggie	9

9. Group the orders by date and calculate the average number of pizzas ordered per day.

```
SELECT
    ROUND(AVG(Pizzas_par_day), 2) AS Avg_num_pizzas
FROM
    (SELECT
        DATE(orders.order_date) AS Order_day,
            SUM(order_details.Quantity) AS Pizzas_par_day
    FROM
        orders
    JOIN order_details ON orders.order_id = order_details.order_id
    GROUP BY DATE(order_date)) AS Daily_Orders;
```

Avg_num_pizzas
138.47

10. Determine the top 3 most ordered pizza types based on revenue

```sql
SELECT
    pizza_types.name,
    ROUND(SUM(order_details.Quantity * pizzas.price),
            2) AS Total_Revenue
FROM
    pizza_types
        JOIN
    pizzas ON pizzas.pizza_type_id = pizza_types.pizza_type_id
        JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY Total_Revenue DESC
LIMIT 3;
```

name	Total_Revenue
The Thai Chicken Pizza	43434.25
The Barbecue Chicken Pizza	42768
The California Chicken Pizza	41409.5

11. Calculate the percentage contribution of each pizza type to total revenue.

```
SELECT
    pizza_types.category,
    ROUND(SUM(order_details.Quantity * pizzas.price) / (SELECT
                    ROUND(SUM(order_details.Quantity * pizzas.price),
                                2) AS Total_sales
                FROM
                    order_details
                        JOIN
                    pizzas ON order_details.pizza_id = pizzas.pizza_id) * 100,
            2) AS Total_Revenue_percentage
FROM
    pizza_types
        JOIN
    pizzas ON pizzas.pizza_type_id = pizza_types.pizza_type_id
        JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY Total_Revenue_percentage DESC;
```

category	Total_Revenue_percentage
Classic	26.91
Supreme	25.46
Chicken	23.96
Veggie	23.68

12. Analyze the cumulative revenue generated over time.

```
SELECT order_date ,
SUM(Revenue) OVER(ORDER BY order_date) AS Cum_Revenue
FROM
(SELECT orders.order_date, ROUND(SUM(order_details.Quantity*pizzas.price),2) AS Revenue
FROM order_details JOIN pizzas
ON order_details.pizza_id = pizzas.pizza_id JOIN orders
ON order_details.order_id = orders.order_id
GROUP BY orders.order_date) AS Sales;
```

13. Determine the top 3 most ordered pizza types based on revenue for each pizza category.

```
SELECT name ,Revenue
FROM
(SELECT category,name,revenue, RANK()
OVER(PARTITION BY category ORDER BY Revenue DESC ) AS rn
FROM
(SELECT pizza_types.category, pizza_types.name ,
ROUND(SUM(order_details.Quantity*pizzas.price),2)AS Revenue FROM pizza_types
JOIN pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id JOIN
order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category, pizza_types.name) AS a) AS b
WHERE rn <=3;
```

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THANK YOU

FOR ATTENTION

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